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Evaluate Yourself

1. What do you mean by fundemantal data types?
2. The data type char is used to represent characters. then why is it often termed as an integer type?
3. What is the advartage of floating point numbers over integers?
4. The data type double is another floating point type. Then why is it treated as a distinct data type?
5. What is the use of void data type?

9.12.2 Memory representation of Fundamental Data types:

One of the most important reason for declaring a variable as a particular data type is to allocate appropriate space in memory. As per the stored program concept, every data should be accommodated in the main memory before they are processed. So, C++ compiler allocates specific memory space for each and every data handled according to the compiler's standards.

The following Table 9.5 shows how much of memory space is allocated for each fundamental data type. Remember that, every data is stored inside the computer memory in the form of binary digits (See Unit I Chapter 2).

Table 9. 5 Memory allocation for Fundamental data types

Data type	Space in memory		Range of value
	in terms of bytes	in terms of bits	
char	1 byte	8 bits	-127 to 128
int	2 bytes	16 bits	-32,768 to 32,767
float	4 bytes	32 bits	3.4×10^{-38} to 3.4×10^{38} -1
double	8 bytes	64 bits	1.7×10^{-308} to 1.7×10^{308}-1

9.12.3 Data type modifiers:

Modifiers are used to modify the storing capacity of a fundamental data type except void type. Usually, every fundamental data type has a fixed range of values to store data items in memory. For example, int data type can store only two bytes of data. In reality, some integer data may have more length and may need more space in memory. In this situation, we should modify the memory space to accommodate large integer values. **Modifiers can be used to modify (expand or reduce) the memory allocation of any fundamental data type. They are also called as Qualifiers.**